Missing Value Treatment and Feature Engineering in Credit Risk Data

Project Overview

This project focused on identifying and treating missing values in a credit risk dataset containing **10,000 observations and 56 variables**. Instead of applying one standard imputation method across all variables, I examined each variable individually to understand *why* the values were missing before deciding how to handle them.

In several cases, missing values did not simply represent incomplete data—they reflected meaningful conditions, such as the absence of a joint applicant, no delinquency history, or no prior credit inquiry. Where appropriate, I preserved this information through indicator variables rather than removing it during cleaning.

As part of this process, I created **6 new flag variables** to capture missingness-related signals, which increased the dataset from **56 to 62 variables**, while reducing missing values to **zero**.

1. Deliverables

This project produced:

- **A full missing value audit across 56 variables**
- **Variable-specific imputation decisions**
- **Six engineered missingness indicators**
- **A cleaned dataset with zero missing values**
- **A documented and reproducible preprocessing workflow**

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2. How Missing Values Were Treated

Debt-to-Income Ratio (debt_to_income)

There were 24 missing values in debt-to-income. After investigating them, I found that 23 occurred when annual income was zero, while one observation was an anomaly and was removed. For the remaining cases, joint debt-to-income information was available.

Treatment

- Removed the anomalous observation
- Created a flag indicating originally missing debt-to-income values
- Filled missing values using debt-to-income values from joint applicants

Why this approach was used

Rather than using a generic statistical fill like a mean or median, I used related information already available in the data. Since the missingness was tied to joint applicants, this approach preserved the logic of the variable better than a blanket imputation method.

Employment Title (emp_title)

Missing employment titles were replaced with **"Unknown."**

Why this approach was used

Because this was a categorical field, assigning "Unknown" preserved all records while keeping missingness explicit rather than silently dropping information.

Joint Annual Income (annual_income_joint)

This variable had 8,505 missing values. Analysis showed those missing values largely represented cases where no joint applicant existed.

Treatment

- Created a **has_joint_applicant** indicator
- Filled missing joint annual income values with zero

Why this approach was used

In this case, the missing values represented absence rather than missing information. Treating them as structural zeros was more appropriate than traditional imputation.

Joint Income Verification (verification_income_joint)

Missing values were replaced with **"Missing."**

Why this approach was used

Here, missingness itself could be informative, so it was preserved as its own category.

Joint Debt-to-Income Ratio (debt_to_income_joint)

Missing values were filled using the borrower's primary debt-to-income ratio.

Why this approach was used

Because missing values appeared where no joint applicant existed, using the primary debt-to-income ratio provided a reasonable substitute based on related borrower information.

Months Since Last Delinquency (months_since_last_delinq)

This variable contained 5,657 missing values. Further analysis showed that 97.3% of these cases corresponded to borrowers with no delinquency history.

Treatment

- Created a **Never_delinqued** indicator

- Filled missing values using **120 months** as a capped sentinel value

Why this approach was used

I treated these missing values as representing borrowers who had never been delinquent, rather than random missing data. I used 120 because it stays within a reasonable range for the variable while clearly separating these cases from observed delinquency durations. The indicator variable preserved this signal separately.

Months Since 90 Days Late (months_since_90d_late)

Treatment

- Created a **Never_been_90days_late** indicator
- Filled missing values with **129**, one greater than the maximum observed value

Why this approach was used

I used max+1 as a practical way to represent "this has never happened." It keeps the value just outside the observed range without introducing an arbitrary outlier, while preserving the ordering of the variable.

Months Since Last Credit Inquiry (months_since_last_credit_inquiry)

This variable had 1,271 missing values.

Treatment

- Created a **Never_have_credit_inquiry** indicator
- Filled missing values with **999**

Why this approach was used

Here I intentionally used a distant sentinel value to separate borrowers with no inquiry history from those with recorded inquiry activity. Since preserving the natural scale was less critical for this variable, this approach provided a clear distinction.

Number of Accounts 120 Days Past Due (num_accounts_120d_past_due)

Treatment

- Created a missingness indicator
- Filled missing values with zero

Why this approach was used

Although observed values were already zero, missingness may still represent incomplete information. Using both a zero fill and an indicator helped preserve that possibility.

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3. Feature Engineering Added

To avoid losing information contained in missingness, I created six additional variables:

1. Debt-to-income missing flag
2. Joint applicant indicator
3. Never delinquent indicator
4. Never been 90 days late indicator
5. No credit inquiry indicator
6. Missing 120-days-past-due indicator

These additions increased the number of variables from **56 to 62**.

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4. Before vs After Summary

Metric	Before	After
Rows	10,000	9,999
Variables	56	62
Missing Values	42191	0
Indicator Variables	0	6

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4. Results

After treatment:

- All missing values were resolved
- Missingness information was preserved through engineered indicators
- Structural missing values were treated differently from random missing values
- The dataset was made ready for downstream analysis and modeling

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5. Conclusion

A key takeaway from this project is that missing values should not always be treated as a simple data quality problem. In several variables, missingness carried information of its own. By analyzing the cause of missingness before imputing values, I was able to improve data quality while also preserving signals that may be useful in modeling.